Proposal: stochastic population dynamics for STREAM

Abstract

The current STREAM model learns a deterministic, autonomous expression-space vector field from optimal-transport-paired developmental snapshots. Across balanced, partial, and unbalanced transport; PCA, KNN, and metacell denoising; adjacent and skipped intervals; and contiguous held-out blocks, causal endpoint prediction remains close to expression persistence. This proposal replaces the deterministic ordinary differential equation with a neural stochastic differential equation and trains predicted populations directly against observed endpoint populations. The drift remains a per-gene, regulatory-sequence-conditioned velocity. A score head and prescribed diffusion represent branching and unresolved fate variation. The recommended first objective is simulation-free score-and-flow matching on noisy stochastic interpolants sampled from an entropic endpoint coupling. A complementary population-level objective directly minimizes endpoint Sinkhorn divergence by reparameterized SDE rollout. We define both objectives, their assumptions, regularization, evaluation, implementation stages, and decision criteria for a focused stochastic-flow ablation.

1 Motivation

The deterministic autonomous model has the form

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_\theta\big(E(x), \{Z_g\}_{g=1}^{G}\big), \tag{1}$$

where $x \in \mathbb{R}^G_{\geq 0}$ is expression, E is the frozen UCE encoder, and Z_g contains the AlphaGenome-derived regulatory tokens linked to gene g. The model predicts one expression velocity per gene. Training uses conditional flow-matching targets formed from optimal-transport pairs between cross-sectional timepoints.

The completed ablations do not establish meaningful improvement over persistence. On the original two-stage holdout, the best mean persistence-normalized Sinkhorn skill is approximately 0.009, while gene-level mean-shift R^2 remains negative. Holding out the middle three stages together increases the best mean Sinkhorn skill to approximately 0.02, but mean-shift R^2 is still negative. A long observed-endpoint chord across the held-out block improves some mean-shift errors without producing a consistent distributional improvement.

An initial score-and-flow implementation with an autonomous flow head and an independent τ-conditioned score head also failed. Validation denoising MSE remained approximately one, the null value for standardized Gaussian noise, under raw, KNN-smoothed, and metacell-smoothed targets. Diffusion improved Sinkhorn distance while worsening gene-level mean shifts, but this could not be attributed to a learned score. This result motivates the coupled conditional-moment parameterization below and an explicit zero-score diffusion control.

These results suggest that the main limitation is not simply model capacity. Cross-sectional OT does not identify a unique descendant for each source cell, development branches, and cell numbers change through proliferation and death. A deterministic map can only assign one local velocity to a state, whereas the data may support a distribution of descendants. Moreover, pairwise velocity regression is not the same objective as accurate endpoint population prediction.

2 Objective and scope

The primary question is whether representing conditional fate variability and training against endpoint distributions improves causal forecasting over persistence. The first stochastic model should therefore:

- retain gene-valued drift and the existing UCE/cCRE conditioning;
- exclude absolute developmental-time labels from the state model;
- compare a strictly autonomous drift with a bridge model conditioned only on the local interpolation coordinate τ;
- avoid requiring one-to-one cross-time cell pairs in the primary loss;
- generate multiple descendants per source cell;
- optimize the same type of endpoint discrepancy used for evaluation;
- separate stochastic biological variation from unrestricted technical noise through a constrained diffusion parameterization.

The initial experiment will not attempt to model division events explicitly, infer lineage trees, or reproduce raw count sampling. A growth-weighted extension is described as a second stage.

3 Gene-scaled dynamical coordinates

Let $q \in \mathbb{R}^G_{>0}$ be fixed per-gene scales estimated from raw training counts, $Q = \mathrm{diag}(q)$, and

$$Y = Q^{-1}X, \qquad X = QY. \tag{2}$$

The implementation performs interpolation, velocity prediction, score matching, and ODE/SDE integration in Y. Before every online UCE call and endpoint metric, it deterministically returns to count-like expression with $X = QY$. OT and train-only PCA remain in X. Because $q_g > 0$, nonnegative projection is equivalent in both coordinate systems.

For a general diffusion,

$$\mathrm{d}Y_t = \mu_Y(Y_t)\mathrm{d}t + G_Y(Y_t)\mathrm{d}W_t \quad \Longleftrightarrow \quad \mathrm{d}X_t = Q\mu_Y(Q^{-1}X_t)\mathrm{d}t + QG_Y(Q^{-1}X_t)\mathrm{d}W_t. \tag{3}$$

Thus the previous gene-scaled Gaussian noise $Q\mathrm{d}W$ becomes isotropic in Y, and an ordinary MSE on Y-velocity is the intended gene-normalized loss in count space. The fixed q is now a checkpointed boundary transform rather than a factor repeated throughout the model equations.

4 Autonomous neural SDE

We propose the expression-space stochastic differential equation

$$\mathrm{d}X_t = \mu_\theta(X_t)\,\mathrm{d}t + G_\phi(X_t)\,\mathrm{d}W_t, \qquad X_t \in \mathbb{R}^G_{\geq 0}, \tag{4}$$

where W_t is an r-dimensional Wiener process. The drift is the existing regulatory model,

$$\mu_{\theta,g}(x) = \mathrm{STREAM}_{\theta,g}(E(x), Z_g)\,. \tag{5}$$

Thus stochastic training does not change the interpretation of the predicted gene velocity.

4.1 Low-rank diffusion

A dense $G \times G$ diffusion matrix is neither identifiable nor tractable for $G = 10{,}000$. Fully independent gene noise is computationally simple but fails to represent coordinated lineage changes. We instead propose a low-rank diffusion with $r = 8$–32:

$$G_\phi(x) = \begin{bmatrix} b_{\phi,1}(E(x), Z_1)^\top \\ \vdots \\ b_{\phi,G}(E(x), Z_G)^\top \end{bmatrix} \in \mathbb{R}^{G \times r}. \tag{6}$$

Each gene's promoter-token representation receives a shared r-dimensional diffusion head. This preserves the shared regulatory architecture and allows one latent noise direction to alter a coordinated set of genes. The diffusion head can use a bounded output,

$$b_{\phi,g} = b_{\max} \tanh(\widetilde{b}_{\phi,g}), \tag{7}$$

which prevents the model from improving a distributional loss by adding unbounded noise.

The lowest-risk first implementation may instead use a state-conditioned scalar $a_\phi(E(x))$ multiplying a learned global $G \times r$ loading matrix. This is less expressive but provides a useful test of whether any coordinated stochasticity helps before adding a second regulatory output head.

5 Differentiable stochastic rollout

For an interval of length $T = t_1 - t_0$, Euler–Maruyama with K steps gives

$$X_{k+1}^{(m)} = \Pi_+ \left[X_k^{(m)} + \mu_\theta(X_k^{(m)})\Delta t + G_\phi(X_k^{(m)})\sqrt{\Delta t}\, \epsilon_k^{(m)} \right], \tag{8}$$

$$\epsilon_k^{(m)} \sim \mathcal{N}(0, I_r), \qquad \Delta t = T/K, \tag{9}$$

where m indexes stochastic particles and Π_+ clamps negative expression values to zero. The Gaussian draws are external random inputs, so Equation 9 is reparameterized and gradients propagate to drift and diffusion parameters.

Online UCE preprocessing includes discrete gene sampling and is frozen. Consequently, the current encoder should be treated as a stop-gradient state map at each step. Gradients still reach earlier updates through the additive identity path in Equation 9, and each step contributes direct parameter gradients. This avoids claiming differentiability through discrete UCE tokenization. Fixed random seeds should be used within each optimization step to reduce gradient variance.

6 Recommended first approach: score-and-flow matching

Simulation-free score-and-flow matching avoids backpropagation through multi-step stochastic rollouts and repeated online UCE evaluations. Given coupled endpoints (x_0, x_1), define the noisy stochastic interpolant

$$X_\tau = (1-\tau)x_0 + \tau x_1 + \gamma(\tau)\Sigma z, \qquad z \sim \mathcal{N}(0, I), \qquad \tau \sim \mathrm{Uniform}(0,1), \tag{10}$$

where $\gamma(0) = \gamma(1) = 0$. One possible schedule is $\gamma(\tau) = \sigma\sqrt{\tau(1-\tau)}$, with endpoint clipping or a smooth bounded alternative used to avoid numerical singularities.

Here Σ is a fixed diagonal matrix of training-only per-gene expression scales. It makes perturbations comparable across genes while retaining gene-valued states and outputs.

Equations 10–17 show the count-space equivalent. With $\Sigma = Q$, the implemented model-coordinate form is

$$Y_\tau = (1-\tau)y_0 + \tau y_1 + \gamma(\tau)z, \tag{11}$$

$$b_Y(y,\tau) = u_Y(y,\tau) + \frac{\dot\gamma(\tau)}{\Delta t} e_\theta(y,\tau), \tag{12}$$

$$s_Y(y,\tau) = -\frac{e_\theta(y,\tau)}{\gamma(\tau)}. \tag{13}$$

The model receives $E(QY_\tau)$, predicts all gene-valued quantities in Y, and uses $X = QY$ only at UCE and evaluation boundaries.

6.1 Coupled conditional-moment heads

The flow and score must not be unrelated readouts. Define two shared τ-conditioned conditional moments,

$$u_\theta(x,\tau) \approx \mathbb{E}[(x_1 - x_0)/\Delta t \mid X_\tau = x], \tag{14}$$

$$e_\theta(x,\tau) \approx \mathbb{E}[z \mid X_\tau = x]. \tag{15}$$

Both are predicted from the same time-modulated promoter features. The same noise estimate e_θ then constructs both fields:

$$b_\theta(x,\tau) = u_\theta(x,\tau) + \frac{\dot\gamma(\tau)}{\Delta t}\Sigma e_\theta(x,\tau), \tag{16}$$

$$s_\theta(x,\tau) = -\Sigma^{-1}\frac{e_\theta(x,\tau)}{\gamma(\tau)}. \tag{17}$$

Thus the noise contribution to the flow and the score cannot disagree without increasing the same noise-regression loss. The objective is

$$\mathcal{L}_{\text{coupled}} = \mathbb{E}\left[\left\|\frac{b_\theta(X_\tau,\tau) - \dot X_\tau}{v_{\text{rms}}}\right\|_2^2 + \lambda_s \left\|e_\theta(X_\tau,\tau) - z\right\|_2^2\right], \tag{18}$$

where $\dot X_\tau = (x_1 - x_0)/\Delta t + \dot\gamma(\tau)\Sigma z/\Delta t$. Predicting standardized noise is equivalent to a $\gamma(\tau)^2$-weighted score loss and avoids endpoint singularities. The normalizer v_{rms} combines the training-only endpoint-velocity RMS with the analytic RMS of the bridge-noise term, preventing intermediate noise levels from dominating the coupled loss.

The same checkpoint contains a strictly autonomous control,

$$b^{\text{aut}}_{\theta,g}(x) = \text{STREAM}^{\text{aut}}_{\theta,g}(E(x), Z_g), \tag{19}$$

trained against $(x_1 - x_0)/\Delta t$ before applying the τ modulation. This isolates the value of the local bridge coordinate from architecture and endpoint samples. Absolute developmental time is supplied to neither path.

6.2 SDE induced by the matched fields

If b_θ satisfies the continuity equation and s_θ estimates the marginal score, then the fields define the SDE

$$\mathrm{d}X_t = \left[b_\theta(X_t,\tau(t)) + \kappa(t)\Sigma^2 s_\theta(X_t,\tau(t))\right]\mathrm{d}t + \sqrt{2\kappa(t)}\Sigma\,\mathrm{d}W_t, \qquad \tau(t) = \frac{t-t_0}{t_1-t_0}. \tag{20}$$

The score correction offsets diffusion in the Fokker–Planck equation, so an exact model has the prescribed interpolant marginals for any nonnegative diffusion schedule κ. Varying κ therefore controls stochasticity without changing the ideal endpoint distribution. Setting $\kappa = 0$ recovers the probability-flow ODE and provides a matched deterministic control. For every positive κ, evaluation also sets the score correction to zero while retaining identical Brownian draws. This distinguishes a learned-score benefit from generic variance broadening.

6.3 Entropic endpoint coupling

Equation 10 still requires a joint endpoint distribution. Independent pairing creates unnecessarily broad paths, while a nearly deterministic OT plan recreates the current arbitrary-pair problem. The recommended compromise is an entropic OT or KL-relaxed UOT coupling. Sampling many endpoint pairs from its nonzero support exposes the model to multiple plausible descendants. This is the simulation-free Schrodinger bridge matching construction described in related work [5, 6].

This approach is simulation-free during fitting: sample an endpoint pair, τ, and z, then regress analytic flow and score targets. It is therefore well suited to frozen, partly discrete online UCE preprocessing. Its main cost is conceptual: the network must receive τ, because bridge noise and drift change along the interpolation. Here τ is a local mathematical path coordinate, not an absolute developmental stage label. Nevertheless, a model conditioned on τ is not strictly autonomous in the sense used by the current no-time-conditioning experiments.

7 Complementary population-level objective

Let $\{x_i^0\}_{i=1}^B$ be source cells and $\{x_j^1\}_{j=1}^{B_1}$ an independently sampled target population. Generate M particles from every source cell and collect the predicted endpoint set

$$\widehat{\mathcal{X}}_1 = \left\{ X_T^{(i,m)} : i = 1, \ldots, B,\ m = 1, \ldots, M \right\}. \tag{21}$$

The principal loss is a debiased Sinkhorn divergence in a train-only, log-normalized PCA scoring space P:

$$\mathcal{L}_{\text{pop}} = S_\varepsilon \Big(P(\widehat{\mathcal{X}}_1), P(\mathcal{X}_1) \Big). \tag{22}$$

PCA is used here as a differentiable population metric and denoising transform, not as the model's dynamical output space. Drift and simulated states remain gene-valued.

This loss aligns training with held-out endpoint evaluation and removes the need to select an individual OT partner for every cell. A minibatch size of $B = 128$–256, $M = 4$ particles, $K = 4$ rollout steps, and rank $r = 8$ is an appropriate initial configuration. Larger M, K, and r should be tested only after establishing a positive endpoint signal.

7.1 Implemented score-flow fine-tuning

The first explicit-rollout implementation starts from the coupled score-and-flow checkpoint rather than introducing a new diffusion head. For each observed training interval it samples source cells and target cells independently. No balanced, partial, or unbalanced OT plan is used to construct cell pairs. Each update jointly evaluates the coupled probability-flow ODE ($\kappa = 0$) and one score-corrected SDE with $\kappa \in \{10^{-3}, 10^{-2}\}$:

$$X_{k+1} = \Pi_+ \Big[X_k + \Delta t \, (v_\theta + \kappa \, \Sigma s_\theta) + \sqrt{2\kappa \Delta t} \, \Sigma \epsilon_k \Big]. \tag{23}$$

Here v_θ and s_θ retain the shared noise parameterization from Equation 20, and Σ is the fixed robust gene scale estimated during simulation-free fitting.

The regulatory trunk, autonomous velocity head, and frozen UCE encoder are held fixed. Fine-tuning updates only the interpolation-coordinate modulation, conditional velocity head, and shared noise head, with a quadratic anchor to their pretrained values. At every integration step, raw current expression is retokenized and encoded by UCE; neither KNN nor metacell smoothing is applied during rollout. The gradient through this discrete state map is truncated, while gradients propagate through the additive expression update and directly into the fields evaluated at each step.

The endpoint objective averages the ODE and SDE losses. Each contains the debiased Sinkhorn divergence in whitened train-only PCA coordinates, a gene-mean penalty normalized by robust gene scale, and a PCA-covariance penalty. The optimized Sinkhorn plan is detached when computing gradients, which applies the envelope gradient without retaining all matrix-scaling iterations. Validation uses disjoint cells, fixed source and target batches, fixed Brownian seeds, and interval-balanced stages. The pretrained checkpoint is update zero, and an exponential moving average of validation endpoint loss selects the fine-tuned checkpoint.

7.2 Regularization

Without constraints, diffusion can absorb model error or produce an overly broad population. We propose

$$\mathcal{L} = \mathcal{L}_{\text{pop}} + \lambda_G \mathbb{E}\left[\|G_\phi(X)\|_F^2\right] + \lambda_\mu \mathbb{E}\left[\|\mu_\theta(X)\|_2^2\right] + \lambda_{\text{mom}}\mathcal{L}_{\text{mom}}, \tag{24}$$

where $\mathcal{L}_{\text{mom}}$ penalizes large errors in endpoint gene means and variances. The moment term is important because the completed deterministic experiments occasionally improve Sinkhorn distance while predicting incorrect gene-level mean shifts. Diffusion should be initialized near zero, and $b_{\max}$ should be calibrated against within-timepoint variation.

To keep the distributional loss informative, target and source samples should be drawn independently at every optimization step. Validation must use fixed cell splits, seeds, source cells, target cells, and Brownian increments.

8 Cheaper paired-transition baseline

Before full rollout training, the existing OT pair banks can support a heteroscedastic transition likelihood:

$$x_1 - x_0 \sim \mathcal{N}(\mu_\theta(x_0)\Delta t, \Sigma_\phi(x_0)\Delta t)\,. \tag{25}$$

For diagonal Σ_ϕ, the negative log likelihood is inexpensive and tests whether state-dependent uncertainty improves calibration. However, this objective inherits arbitrary OT pairing and can reward inflated variance. It should be treated as an implementation smoke test and baseline, not the main stochastic-flow result.

9 Other alternatives

Iterative Schrodinger bridge fitting. Iterative proportional fitting can learn forward and backward stochastic bridges between endpoint distributions [4]. This is a principled distributional approach but requires repeated simulation and time-dependent score or drift models. It should be considered if the simpler autonomous SDE shows a signal.

Independent per-gene noise. A diagonal diffusion head is easy to implement but mostly represents technical noise and cannot express correlated fate decisions. It remains a useful negative control against low-rank diffusion.

10 Growth and changing population mass

An SDE transports probability mass but does not create or remove it. If proliferation and death are central, add a growth-rate head $\gamma_\psi(E(x))$ and particle weights

$$w_{k+1} = w_k \exp(\gamma_\psi(X_k)\Delta t)\,. \tag{26}$$

Training then uses weighted or unbalanced Sinkhorn divergence. This separates state motion, stochastic fate dispersion, and mass change. Growth should be introduced only after comparing deterministic and stochastic transport under fixed mass; otherwise diffusion and growth may compensate for each other and be difficult to identify.

11 Evaluation

For each source cell, generate $M_{\text{eval}} = 16$–64 particles with fixed seeds. Report:

- debiased Sinkhorn divergence and skill relative to persistence;
- gene-level mean-shift R^2 and mean absolute error;
- mean per-gene Wasserstein-1 distance;
- endpoint covariance error in train-only PCA space;
- coverage and calibration of observed cells under generated particles;
- metrics stratified by annotated cell type or lineage where available;
- the autonomous ODE and coupled probability-flow ODE from the same checkpoint;
- matched stochastic rollouts with learned versus zero score correction and identical Brownian seeds;
- sensitivity to particle count and integration steps.

Persistence should be represented by repeated source cells, so it has the same number of particles as the stochastic model. A deterministic STREAM model, the same SDE with $G = 0$, and a pure-diffusion model with $\mu = 0$ provide the necessary ablations. Pairwise displacement R^2 should remain secondary because no individual descendant is observed.

12 Experimental plan

The first full comparison should use the contiguous middle-three-stage holdout. Train separate adjacent, skip-1, and skip-2 models only if Stage 1 beats persistence. The observed bridge interval across the block can be added as a subsequent controlled factor. Because Stage 1 is time-dependent while Stage 3 is autonomous, conclusions must distinguish improved stochastic modeling from the additional interpolation-coordinate input.

13 Implementation plan

The implementation should preserve package boundaries:

1. Add an autonomous velocity readout and shared τ-conditioned velocity/noise readouts to the reusable model package. Construct flow and score analytically from the same noise prediction. Keep current drift-only checkpoints backward compatible.

Table 1: Proposed staged experiments. Advancement requires improved held-out population metrics without loss of gene-level mean-shift accuracy.

Stage	Model	Training objective	Decision
0	Existing deterministic drift	Existing CFM loss	Reproduce current checkpoint and evaluation
1	Score-and-flow STREAM	Entropic-coupling flow and denoising-score regression	Verify analytic targets and endpoint marginals
2	Matched SDE and ODE	Vary κ; disable score at inference	Require positive skill and a specific stochastic benefit
3	Global rank-8 autonomous diffusion	Population Sinkhorn, $M = 4$, $K = 4$	Compare simulation-free and rollout-trained objectives
4	CRE-conditioned rank-8 diffusion	Population Sinkhorn plus moment loss	Test state/gene-specific uncertainty
5	Optional growth head	Weighted unbalanced Sinkhorn	Proceed only if mass mismatch remains limiting

2. Add a stochastic-interpolant sampler returning entropically coupled endpoints, τ, Gaussian noise, and analytic flow/score targets.
3. Add the coupled flow/noise loss, with tests against analytic Gaussian bridges, known marginal scores, and the zero-score SDE control.
4. Add a reusable SDE rollout for Equation 20, plus a probability-flow ODE mode with $\kappa = 0$.
5. Add an autonomous drift–diffusion output path and a reusable Euler–Maruyama rollout accepting fixed Gaussian noise, particle count, diffusion rank, and nonnegativity projection.
6. Add differentiable train-only PCA scoring and debiased Sinkhorn loss in torch. Do not route model outputs through PCA.
7. Add an interval-stratified population sampler that returns independent source and target cells rather than OT pairs.
8. Add fixed stochastic validation batches containing cell indices and Brownian seeds. Select checkpoints by endpoint Sinkhorn EMA, with moment metrics as safeguards.
9. Extend causal evaluation to generate multiple particles, report stochastic calibration, and retain deterministic/persistence baselines.
10. Run synthetic tests where a branching SDE has known drift and diffusion before fitting atlas data.

The highest-risk engineering component is repeated online UCE evaluation for $B \times M$ particles. Initial rollout training should cache repeated source encodings where possible, use small M and K, and profile whether UCE or the gene-conditioned drift dominates runtime. A lower-cost expression-state SDE can validate the training objective before enabling online UCE.

14 Success and stopping criteria

The stochastic model should be considered useful only if it satisfies all of the following on held-out stages:

- Sinkhorn skill is consistently positive across horizons and materially larger than the current one-to-two percent range;
- mean-shift R^2 is nonnegative or improves without a compensating degradation in Wasserstein distance;
- performance is stable across Brownian seeds and particle counts;
- learned diffusion is nonzero, bounded, and improves over both drift-only and pure-diffusion controls;
- improvement persists within major cell types rather than arising only from global mixture smoothing.

If population-level SDE training still matches persistence, the appropriate conclusion is that expression snapshots and the present state representation do not identify the requested dynamics at this resolution. Further architecture sweeps would then be lower priority than lineage constraints, explicit cell types, perturbations, or direct proliferation information.

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